

Date	29 June 2026
Team ID	SWTID-2026-7290
Project Name	Smart Lender
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Bank officers securely log in using a username and password before accessing the application.	High
2	Authorization Levels	Only authorized bank staff can access applicant records and prediction results. Administrators can manage users and models.	High
3	External Interfaces	Integrates with the trained XGBoost model and optionally IBM Cloud services for deployment and storage.	High
4	Transaction Processing	Accept applicant details, validate input data, generate loan approval predictions, and store prediction history.	High
5	Reporting	Display prediction results and maintain a history of loan evaluations for future analysis.	Medium
6	Business Rules	Loan decisions are based on applicant details such as income, credit history, education, employment status, loan amount, and property area using the trained ML model.	High
7	Compliance to Laws or Regulations	Ensure secure handling of applicant data and comply with applicable banking data privacy regulations.	High
8	Other	Allow administrators to update or replace the trained machine learning model without affecting the user interface.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Generate loan prediction quickly after user submission.	Response time < 2 seconds
2	Scalability	Support increasing numbers of users and loan applications.	Handle at least 500 concurrent users
3	Security & Data Privacy	Protect applicant information using secure communication and encrypted storage where applicable.	HTTPS enabled and secure data handling
4	Reliability & Availability	Ensure consistent availability of the application.	99.9% uptime
5	Usability & Accessibility	Provide a responsive, easy-to-use interface for bank staff.	Compatible with modern browsers and mobile devices
6	Maintainability	Modular architecture for easy updates to the ML model and application.	Code follows modular design principles